Riya Panche

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EDUCATION

University of California, Irvine **Irvine, CA**
B.S. in Computer Science and Engineering; B.S. in Electrical Engineering *June 2028*

- **Relevant Coursework**: Embedded Systems, Data Structures and Algorithms, Network Analysis, Concepts in Programming Languages, Verilog, Digital Systems, Differential Equations, Physics: Mechanics and Electromagnetism, Discrete Mathematics

TECHNICAL SKILLS

- **Languages**: Python, C, C++, Java, JavaScript, SQL, Verilog, MATLAB, R, HTML, CSS
- **Embedded & Hardware**: STM32 (bare-metal C), SPI, I2C, UART, PWM, ADC, KiCad, Fusion 360, AutoCAD, COMSOL, OpenRocket
- **Software & Tools**: Django, React, Flask, PostgreSQL, SQLite, Docker, Microsoft Azure, Linux/Unix, Visual Studio Code, PyCharm, IntelliJ, Google Colab

EXPERIENCE

VenuAI **Remote**
Software Engineering Intern *July 2026 - September 2026*

- Developed and shipped full-stack features using Python, Django, React, Microsoft Azure, and PostgreSQL, addressing product pain points identified through direct user testing
- Diagnosed and resolved production issues across backend APIs, frontend integrations, authentication, CORS, and deployment
- Collaborated in a fast-paced startup environment to rapidly iterate on features and deliver improvements directly to customers

Magnetic Microsystems & Microrobotics Lab **UC Irvine**
Undergraduate Research Assistant *April 2025 - Present*

- Developed a real-time Flask dashboard to visualize TLE493D magnetic sensor data, enabling live monitoring and post-test analysis of sensor performance
- Engineered a custom GUI for magnetic field visualization, integrating robotic arm telemetry to enable precise 40° arc magnet tracking and real-time data analysis
- Programmed a MyCobot robotic arm to sweep an arc and track magnet position from live sensor data
- Developing an autonomous WPT docking robot using magnetic sensing (TLE493D/SPI), TinyZero, and an L298N motor driver

PROJECTS

Splitwise 2.0 - Receipt Splitting PWA - *Personal project* | https://splitwise2.onrender.com/

- Built a full-stack PWA that extracts itemized receipt data from photos and calculates per-person splits (equal, percentage, exact-amount)
- Fine-tuned a Donut vision-transformer model on CORD-v2 to parse receipt images directly into structured JSON, removing the need for a separate OCR step
- Designed a Strategy + Factory pattern split-calculation engine (Flask/SQLite) allowing new split types to be added without touching existing logic
- Containerized with Docker and deployed to production on Render, confirmed working as an installable mobile PWA

Avionics Engineer, Drag Sail Deorbit Project - *SEDS at UC Irvine* *October 2026 - Present*

- Contribute to embedded systems development for a student-led spacecraft drag sail deorbit project focused on space sustainability
- Work within the Avionics - Embedded subteam on embedded software and hardware interfacing for spacecraft avionics
- Develop and integrate embedded software that interfaces directly with the avionics hardware of the deorbit system
- Collaborate with multidisciplinary engineering subteams to support avionics integration and overall mission objectives

Bare-Metal STM32 Embedded Firmware - *STM32H563ZI Nucleo* *Spring 2026*

- Developed firmware in bare-metal C for an STM32H563ZI Nucleo board using direct register-level access (no HAL), covering GPIO, interrupts, timers, PWM, ADC, and I2C peripherals
- Built a tone player/recorder with a 4x4 keypad, speaker, mode LEDs, and I2C EEPROM, and a multiplication calculator on a 16x2 LCD driven by a five-state machine

Micromouse - *IEEE at UC Irvine* *2026 - 2027*

- Developing an autonomous maze-solving robot in a 3-person team through UCI's IEEE Micromouse program, covering custom PCB design, STM32 firmware in C/C++, and floodfill pathfinding

LEADERSHIP EXPERIENCE

Business Director - SEDS at UC Irvine *October 2026 - Present*

- Develop sponsorship proposals, outreach communications, and partnership materials to engage companies and secure funding and resources for SEDS at UCI
- Support sponsorship strategy, fundraising initiatives, and financial planning to sustain club operations and student-led aerospace projects

VenusHacks Organizer *October 2025 - May 2026*

- Assist with corporate outreach by contacting 70+ potential sponsors and partners to secure funding, mentoring, and in-kind support
- Co-organized end-to-end logistics for a women-centric hackathon serving 100+ participants, coordinating venue, scheduling, and sponsor relations